Template for Enrichment task 2: Programming the RSA algorithm

Question 1: Th RSA algorithm

In [24]:
```
# Function definitions

def rsafunction1(p, q):
    n = p * q
    phi = (p - 1) * (q - 1)

    e = 7
    while gcd(e, phi) != 1:
        e = next_prime(e)

    d = inverse_mod(e, phi)

    return [n, e, d]


def rsafunction2(m, e, n):
    c = power_mod(m, e, n)
    return c


def rsafunction3(c, d, n):
    m = power_mod(c, d, n)
    return m
```

In [25]:
```
# RSA Function 1 testing

rsafunction1(17, 11)
```

Out[25]: [187, 7, 23]

In [26]:
```
# RSA Function 2 testing


rsafunction2(88, 7, 187)
```

Out[26]: 11 In [27]:

```
# RSA Function 3 testing

rsafunction3(11, 23, 187)
```

Out[27]: 88

In [28]:

```
# Question 2

# Exercise 5.1.1 (b)

m = 1234
e = 17
n = 9797

# c = rsafunction2(m, e, n)

rsafunction2(m,e,n)
```

Out[28]: 8897